Fanions at every rank: finite budgets, hybrid crossovers, and Nesterov tracking

Second research attempt for the Neon repository

21 October 2026

Abstract

We extend the comparison to every rank k in the plain, F-, and S-Fanion families. A single convergence theorem retains their different support functions, gradient spectra, entry density, cumulative learning rates, and oracle errors. Its momentum corollary uses the repository's actual Nesterov convention. We strengthen the quadratic lower bounds by retaining the whole initial singular spectrum and by separating a hybrid's rank-limited and dense updates. An exactly solvable spectral flow explains both $1/k$ progress and its saturation; hybrid examples identify different crossover ranks. The saved quadratic traces provide unusually close evidence for the low-rank time scale: Fanion-2 and Fanion-10 nearly coincide with Neon after rescaling time by k. A source audit also finds a different learning-rate schedule for F-Fanion-10. The results provide a substantial explanation for the quadratic experiments. They do not yet prove the CIFAR/LLM accuracy ordering.

What changed in this attempt

The first note, `bounds.pdf`, established rank and nuclear-budget obstructions and a gradient/curvature score. This note develops the missing *rank curve*, including all intermediate Fanions and both hybrid families. All code and outputs are in `theory_attempts`; the submission and benchmark code have not been modified.

Three useful conclusions.

1. At the same small learning rate, low-rank Fanions spend nuclear update budget at rate $k\gamma$. This predicts approximately k times faster progress while that budget is the binding obstruction. It is exceptionally consistent with the saved rank-2 and rank-10 quadratic trajectories.
2. F- and S-Fanions supply a dense update as well. Their relevant support functions are $aS_k + b\|G\|_{\mathrm{F}}$ and $aS_k + b\eta\|G\|_{1,\mathrm{entry}}$, respectively. The extra terms explain why low-rank hybrids can behave very differently from plain Fanions, and why increasing their rank need not give the same k-fold gain.
3. A complete ordering needs spectral *shape*, curvature, and momentum alignment. Three gradient norms determine the endpoint comparison but do not determine the intermediate ranks, even in a noiseless isotropic quadratic. We give an explicit counterexample and useful intervals.

A concrete experimental confound.

In the current `code/L_smooth_tests/run_benchmarks_new.py`, F-Fanion-10 alone uses `inv_sqrt_lr`; its other Fanion specifications use `const_lr`. Over 5,000 updates, its cumulative learning rate is 2.09952 rather than 75. The slow F-Fanion-10 trace therefore cannot be used as a comparison of ranks at a common schedule. The CSV itself does not serialize configuration; this diagnosis uses the current source and its consistency with the saved trace, rather than claiming a recovered run manifest.

One convergence bound for all three families

Let $r = \min(m, n)$, $d = mn$, $1 \le k \le r$, and let singular values be nonincreasing. Define $S_k(H) = \sum_{i=1}^{k} \sigma_i(H)$ and the partial polar direction $D_k(H) = \sum_{i=1}^{k} u_i v_i^{\top}$, omitting zero singular directions. Take $a, b \ge 0$; the experiments use $a = b = 1/2$. The ideal directions are

$$\begin{aligned} D_k^{\mathrm{P}}(H) &= D_k(H), \\ D_k^{\mathrm{F}}(H) &= aD_k(H) + bH/\|H\|_{\mathrm{F}}, \\ D_k^{\mathrm{S}}(H) &= aD_k(H) + b\eta \operatorname{sign}(H). \end{aligned} \tag{1}$$

All directions are defined as zero at $H = 0$. They maximize the linear functional $\langle H, D \rangle$ over the corresponding symmetric convex sets

$$C_k = \{D : \|D\|_{\mathrm{op}} \le 1,\ \|D\|_* \le k\}, \qquad C_k^{\mathrm{F}} = aC_k + bB_{\mathrm{F}}, \qquad C_k^{\mathrm{S}} = aC_k + b\eta B_{\infty}^{\mathrm{entry}}.$$

When $\operatorname{rank}(H) \ge k$, the partial polar maximizer is an extreme choice in C_k; other maximizers can occur at tied singular values. The support functions ϕ_k and safe Frobenius radii R_k are:

Family	$\phi_k(G)$	R_k
Plain	$S_k(G)$	$\sqrt{k}$
F	$aS_k(G) + b\|G\|_{\mathrm{F}}$	$a\sqrt{k} + b$
S	$aS_k(G) + b\eta\|G\|_{1,\mathrm{entry}}$	$a\sqrt{k} + b\eta\sqrt{d}$

The radius inequalities use the triangle inequality and need not be sharp.

A broad spectrum without a condition number assumption.

For $G \ne 0$, put

$$\theta(G) = \frac{\|G\|_*}{\sqrt{r}\,\|G\|_{\mathrm{F}}}, \qquad c(G) = \frac{\|G\|_{1,\mathrm{entry}}}{\sqrt{d}\,\|G\|_{\mathrm{F}}}.$$

Here $\theta^2 r$ is the nuclear effective rank, and c measures entry density and uniformity. Since $S_k/k \ge \|G\|_*/r$, a lower bound $\theta(G_t) \ge \theta_0 > 0$ implies a simultaneous bound for *all* k. If also $c(G_t) \ge c_0 > 0$ when needed, then

$$\begin{aligned} \phi_k(G_t) &\ge \ell_k \|G_t\|_{\mathrm{F}}, \\ \ell_k^{\mathrm{P}} &= \theta_0 k/\sqrt{r}, \\ \ell_k^{\mathrm{F}} &= a\theta_0 k/\sqrt{r} + b, \\ \ell_k^{\mathrm{S}} &= a\theta_0 k/\sqrt{r} + b\eta c_0\sqrt{d}. \end{aligned} \tag{2}$$

These assumptions concern the true gradients along the particular trajectory. A median from logged minibatch gradients does not establish a uniform lower bound for an unobserved trajectory.

> **Theorem 1** (All ranks, schedules, and hybrid families). *Suppose f is Frobenius L-smooth and bounded below by f_*. Write $G_t = \nabla f(X_t)$, $\Delta = f(X_0) - f_*$, and $X_{t+1} = X_t - \gamma_t \tilde{D}_t$, with $\gamma_t \ge 0$. For one of the three sets above suppose $\tilde{D}_t$ is feasible, $\|\tilde{D}_t\|_{\mathrm{F}} \le R_k$, and*
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> $$\delta_t = \phi_k(H_t) - \langle H_t, \tilde{D}_t \rangle \ge 0.$$

The input H_t may first be positively rescaled; compute both δ_t and $E_t = H_t - G_t$ from that input. If $\phi_k(G_t) \geq \ell_k \|G_t\|_{\mathrm{F}}$ with $\ell_k > 0$, then, for $\Gamma_T = \sum_{t<T} \gamma_t > 0$,

$$\frac{\sum_{t<T} \gamma_t \|G_t\|_{\mathrm{F}}}{\Gamma_T} \leq \frac{\Delta + \frac{L}{2} R_k^2 \sum_{t<T} \gamma_t^2 + 2 \sum_{t<T} \gamma_t \phi_k(E_t) + \sum_{t<T} \gamma_t \delta_t}{\ell_k \Gamma_T}. \tag{3}$$

For stochastic iterates, the expected version holds by taking expectations, provided the spectral/density conditions hold almost surely.

Proof. The support function is a norm, so feasibility and its triangle inequality give

$$\langle G_t, \tilde{D}_t \rangle \geq \phi_k(H_t) - \delta_t - \phi_k(E_t) \geq \phi_k(G_t) - 2\phi_k(E_t) - \delta_t.$$

Insert this into the descent lemma, use the radius bound, and telescope. □

Corollary 2 (Explicit rank dependence). *For $L, \Delta, R_k > 0$, exact gradients, and exact oracles with constant $\gamma = \sqrt{2\Delta/(LR_k^2 T)}$,*

$$\frac{1}{T} \sum_{t<T} \|G_t\|_{\mathrm{F}} \leq \frac{R_k}{\ell_k} \sqrt{\frac{2L\Delta}{T}}. \tag{4}$$

Consequently $T \geq 2L\Delta R_k^2/(\ell_k^2 \varepsilon^2)$ suffices for the average gradient norm to be at most ε. For plain Fanions this factor is

$$K_k := R_k^2/\ell_k^2 = \frac{r}{\theta_0^2 k}.$$

At a common fixed γ, their bound instead reads

$$\frac{1}{T} \sum_{t<T} \|G_t\|_{\mathrm{F}} \leq \frac{\sqrt{r}}{\theta_0} \left(\frac{\Delta}{k\gamma T} + \frac{L\gamma}{2} \right). \tag{5}$$

Thus the same asymptotic exponent can conceal a factor r/k in the deterministic iteration guarantee. Equation (5) is especially relevant to inherited learning rates: increasing k reduces the transient term by k, while the displayed residual term is independent of k. Comparing upper bounds is not a proof that two trajectories attain them or that the methods are universally ordered.

What the hybrid constants predict.

For F-Fanions, the support terms become comparable at $k \approx (b/a)\sqrt{r}/\theta_0$. Below this scale, the normalized-gradient part can dominate the guaranteed projection even though it has a small coefficient. For S-Fanions the analogous scale is $k \approx (b/a)\eta c_0 \sqrt{rd}/\theta_0$. The curvature radii also change with k, so these projection crossovers are not an assertion that the optimally tuned rank curve must be monotone.

Tracking the code's actual Nesterov input

The code uses $B_{-1} = 0$ and

$$B_t = \beta B_{t-1} + g_t, \qquad H_t = g_t + \beta B_t.$$

This differs from placing a factor $1-\beta$ inside the buffer update. The ideal oracles in (1) are invariant to positive scaling, so take $\widehat{H}_t = (1-\beta)H_t$. Put $\alpha = 1-\beta$. Its exact weights on past stochastic gradients are

$$\widehat{H}_t = \sum_{j=0}^{t} w_j g_{t-j}, \qquad w_0 = 1-\beta^2, \quad w_j = \alpha\beta^{j+1}\ (j \geq 1), \quad \sum_{j=0}^{t} w_j = 1-\beta^{t+2}. \tag{6}$$

The missing initial mass must be retained in a tracking proof.

Proposition 3 (Nesterov tracking for every Fanion). *Assume constant γ, $0 \leq \beta < 1$, the radius bound R_k, and $g_t = G_t + \xi_t$, where ξ_t are martingale differences with $\mathbb{E}\left\|\xi_t\right\|_{\mathrm{F}}^2 \leq \sigma^2$. For deterministic X_0,*

$$\mathbb{E}\left\|\widehat{H}_t - G_t\right\|_{\mathrm{F}} \leq \beta^{t+2}\left\|G_0\right\|_{\mathrm{F}} + \frac{\beta^2}{\alpha}L\gamma R_k + \sigma\sqrt{v_\beta}, \qquad v_\beta = (1-\beta^2)^2 + \frac{\alpha\beta^4}{1+\beta} \leq 2\alpha. \tag{7}$$

For exact oracles the expected average gradient norm is bounded by

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\left\|G_t\right\|_{\mathrm{F}} \leq \frac{1}{\ell_k}\left\{\frac{\Delta}{\gamma T} + L\gamma R_k^2\left(\frac{1}{2} + \frac{2\beta^2}{\alpha}\right) + \frac{2R_k\beta^2\left\|G_0\right\|_{\mathrm{F}}}{\alpha T} + 2R_k\sigma\sqrt{v_\beta}\right\}. \tag{8}$$

Proof. Unroll (6) and separate the noise, the missing initial gradient, and gradient differences. Every iterate displacement has norm at most γR_k. The total coefficient for its drift is

$$\sum_{j=0}^{t} j w_j + t\beta^{t+2} = \frac{\beta^2(1-\beta^t)}{\alpha} \leq \frac{\beta^2}{\alpha}.$$

The squared coefficients of the martingale differences sum to at most $\sum_{j\geq 0} w_j^2 = v_\beta$; orthogonality and Jensen's inequality bound their expected norm by $\sigma\sqrt{v_\beta}$. Finally $\phi_k(E) \leq R_k\|E\|_{\mathrm{F}}$ and $\sum_{t<T}\beta^{t+2} \leq \beta^2/\alpha$. Apply Theorem 1 to $\widehat{H}_t$. □

For nonzero noise, taking α of order $\min\{1, \sqrt{L\Delta/(\sigma^2 T)}\}$ and γ of order $\sqrt{\Delta\alpha/(LR_k^2 T)}$ yields, apart from the explicit initialization term, the usual leading dependence

$$O\left(\frac{R_k}{\ell_k}\left[\sqrt{\frac{L\Delta}{T}} + \sqrt{\sigma}\left(\frac{L\Delta}{T}\right)^{1/4}\right]\right).$$

In the stochastic large-T regime this corresponds to a leading sufficient iteration count proportional to $K_k^2 L\Delta\sigma^2/\varepsilon^4$. For broad-spectrum plain Fanions, $K_k^2 = r^2/(\theta_0^4 k^2)$. The exponent remains shared, but the rank dependence does not disappear. Using Frobenius noise here is conservative; (3) can use $S_k(E)$, $\|E\|_{\mathrm{F}}$, and $\|E\|_{1,\mathrm{entry}}$ separately if measured or controlled.

Implementation scope.

These results cover the ideal directions and feasible approximate oracles with a recorded support deficit. The repository's finite Newton–Schulz polynomial need not satisfy $\|D\|_{\mathrm{op}} \leq 1$ exactly. Its actual norm and projection error must be inserted before using this theorem for practical Muon. The tiny denominator

regularizer in normalized-gradient code is also an oracle approximation. Theorems assume additive updates; weight normalization or retractions require a separate calculation.

A stronger finite-budget obstruction

For the paper's quadratic let

$$f(X) = \frac{1}{2}\left\| M^{1/2}(X - Z)N^{1/2} \right\|_{\mathrm{F}}^2, \quad E_0 = M^{1/2}(X_0 - Z)N^{1/2}, \quad L = \|M\|_{\mathrm{op}}\|N\|_{\mathrm{op}}, \quad s_i = \sigma_i(E_0).$$

For $0 \le p \le r$, $P, B \ge 0$, define

$$\Psi(s; p, P, B) = \frac{1}{2}\sum_{i=1}^{r} (s_i - a_i)^2, \qquad a_i = \begin{cases} \min\{P, (s_i - \tau)_+\}, & i \le p, \\ 0, & i > p. \end{cases} \tag{9}$$

Take $\tau = 0$ if $B \ge \sum_{i \le p} \min(P, s_i)$; otherwise choose it so $\sum_i a_i = B$. For $p = 0$ or $PB = 0$, all a_i are zero.

> **Theorem 4** (Singular-budget lower bound). *Among all matrices A with $\mathrm{rank}(A) \le p$, $\|A\|_{\mathrm{op}} \le P$, $\|A\|_* \le B$, the minimum of $\left\|E_0 - A\right\|_{\mathrm{F}}^2/2$ equals $\Psi(s; p, P, B)$. Therefore any additive plain Fanion-k trajectory, with arbitrary inputs, momentum, and nonnegative step sizes, satisfies*
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> $$f(X_T) \ge \Psi(s; \min(kT, r), \sqrt{L}\,\Gamma_T, \sqrt{L}\,k\Gamma_T). \tag{10}$$

Proof. Von Neumann's trace inequality shows that the best correction aligns its singular vectors with those of E_0. Its nonnegative singular corrections can be restricted to the first p indices and to $a_i \le s_i$. The remaining scalar problem minimizes $\sum_{i \le p} (s_i - a_i)^2/2$ subject to $0 \le a_i \le P$ and $\sum a_i \le B$. Its optimality conditions give the clipped threshold in (9). The result is sorted, so it is a feasible singular spectrum. For the trajectory, the whitened cumulative correction has rank at most kT, operator norm at most $\sqrt{L}\,\Gamma_T$, and nuclear norm at most $\sqrt{L}\,k\Gamma_T$. Apply the minimization result. □

This lower bound simultaneously includes the earlier rank-tail and nuclear bounds. It is exact for the relaxed correction constraints; it does not say that any optimizer can realize every admissible correction in T steps. The operator budget matters at high k: serving all modes at once cannot remove the largest singular error faster than the allowed per-mode step.

Hybrids retain a limited rank component.

Let $R_{\mathrm{dense}} = 1$ for F-Fanions and $R_{\mathrm{dense}} = \eta\sqrt{d}$ for S-Fanions. Decompose their accumulated update into its partial-polar and dense parts. Set

$$Q_T = \Psi(s; \min(kT, r), a\sqrt{L}\,\Gamma_T, ak\sqrt{L}\,\Gamma_T).$$

The dense part has whitened Frobenius norm at most $bR_{\mathrm{dense}}\sqrt{L}\,\Gamma_T$. The triangle inequality gives

$$f(X_T) \ge \frac{1}{2}\left[\sqrt{2Q_T} - bR_{\mathrm{dense}}\sqrt{L}\,\Gamma_T\right]_+^2. \tag{11}$$

In addition the total correction has operator budget $(a + bR_{\mathrm{dense}})\sqrt{L}\,\Gamma_T$ and nuclear budget $(ak + b\sqrt{r}\,R_{\mathrm{dense}})\sqrt{L}\,\Gamma_T$. Thus another valid lower bound is

$$f(X_T) \geq \Psi(s; r, (a + bR_{\text{dense}})\sqrt{L}\,\Gamma_T, (ak + b\sqrt{r}\,R_{\text{dense}})\sqrt{L}\,\Gamma_T). \tag{12}$$

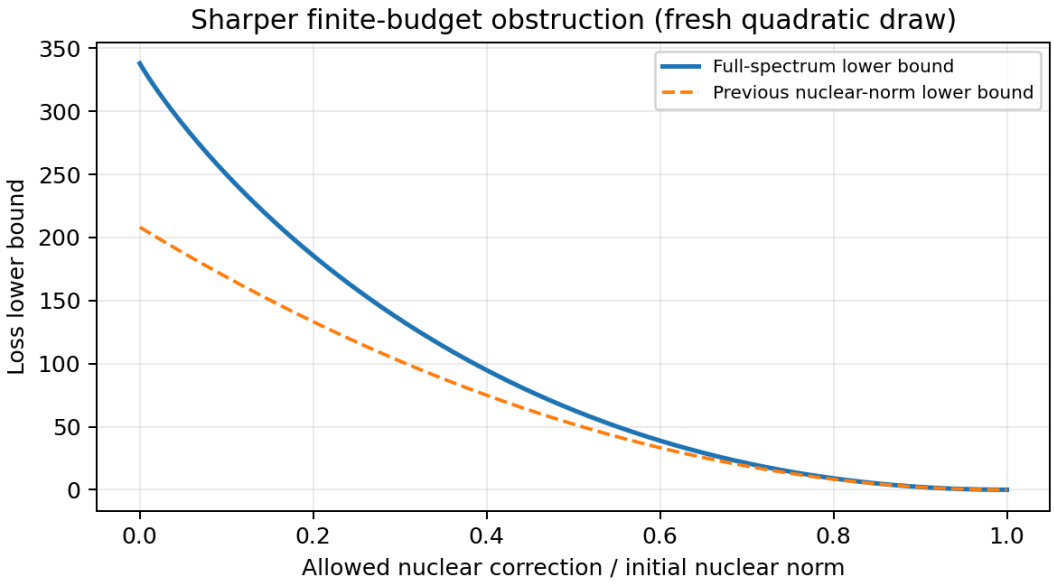
We use the maximum of (11) and (12). It is incorrect to impose $\operatorname{rank}(X_T - X_0) \leq kT$ on a hybrid's *whole* update; the dense part may be full rank from its first step.

Numerical consequence for the paper's problem distribution.

Using the same fresh NumPy seed 142 draw as the first note, $r = 500$, $f(X_0) = 337.9335$, $\|E_0\|_* = 456.2072$, and $\sqrt{L} = 0.997828$, the following are necessary iteration counts for loss 10^{-3}. They are lower bounds, not achieved training times, and the draw is not the original PyTorch seed 42.

k	**Plain,** $\gamma = .007$	**F,** $\gamma = .015$	**S,** $\gamma = .011$
1	65,172	3,197	929
2	32,586	2,990	913
10	6,518	2,056	816
100	652	501	412
500	392	279	258

Here $a = b = 1/2, \eta = .01$. Retaining the operator budget raises the ideal Muon obstruction from 131 to 392 steps; the split bound raises F-Neon's from 2,604 to 3,197. These remain conservative necessary times. With the F-Fanion-10 inverse-square-root schedule, the necessary time from the same constraints becomes 1,056,970 steps. For this last calculation we use the safe upper bound $\Gamma_T \leq .015(2\sqrt{T} - 1), T \geq 1$.



Retaining the spectrum strengthens the earlier scalar nuclear bound. This figure only displays the nuclear constraint; operator and rank constraints can strengthen it further. Fresh draw, not saved training.

Solvable rank dynamics and hybrid crossover scales

An exact finite-step Fanion example

Take $f(X) = \|X\|_{\mathrm{F}}^2/2$, $X_0 = sI_r$, and constant $\gamma = s/h$ with h a positive integer. Select the top k positive residual singular directions each step, stopping at zero. Let $n_T = \min(kT, rh) = qr + p$, $0 \le p < r$. Then

$$f(X_T) = \tfrac{1}{2}[(r-p)(s-q\gamma)^2 + p(s-(q+1)\gamma)^2]. \tag{13}$$

Indeed the k largest residuals are the least-served coordinates, so their service counts differ by at most one. Before extinction,

$$f(X_T) = \frac{r}{2}\left(s - \frac{k\gamma T}{r}\right)^2 + \frac{\gamma^2 p}{2}\left(1 - \frac{p}{r}\right). \tag{14}$$

The rounding remainder is at most $r\gamma^2/8$. This is an actual discrete Fanion trajectory and proves an approximately exact dependence on kT, with extinction after $\lceil rh/k \rceil$ steps. Momentum and variable curvature are not part of this example.

A whole-spectrum continuous model

For the isotropic quadratic, consider the convexified spectral flow $\dot{X} \in -\partial S_k(X)$. At a tie, this allows averages of tied rank-k maximizers. If $X_0 = U\mathrm{diag}(s)V^{\top}$, its solution is

$$X(u) = U\mathrm{diag}(s_i - a_i(u))V^{\top}, \qquad a_i(u) = \min\{u, (s_i - \tau(u))_+\}, \tag{15}$$

where $\sum_i a_i(u) = \min\{ku, \sum_i \min(u, s_i)\}$. Its exact extinction time is

$$u_*^{\mathrm{P}} = \max\left\{s_1, \frac{\sum_i s_i}{k}\right\}. \tag{16}$$

Proof. The formula is continuous and piecewise affine. On an interval with a positive threshold, let c corrections be capped at u and let j be strictly between zero and u. Differentiating their sum ku gives rate one on the capped modes and rate $(k-c)/j$ on the tied residual plateau. If $j = 0$, then $c = k$ and only the capped modes move, at rate one. This rate lies in $[0, 1]$; the selected modes are the largest residuals, and the total rate is k. Thus the negative derivative is in $\partial S_k(X)$. When the nuclear constraint is inactive, every remaining positive residual decreases at rate one; their number is at most k. The expression is zero exactly when both $u \ge s_1$ and $ku \ge \sum_i s_i$. Uniqueness follows from monotonicity of the subdifferential of the convex function S_k. □

The crossover is $k \approx \|X_0\|_*/\|X_0\|_{\mathrm{op}}$: below it the nuclear budget gives $1/k$ improvement; above it the largest mode limits progress. This is more informative than a uniform r/k slogan. The continuous model permits full-rank averages at ties, so (15) is not an exact finite-step formula for a literal rank-k implementation. It is a convexified small-step model. Equation (13) supplies a finite-step check in an important special case.

F-Fanion flow is a time change of the plain flow

For $a > 0$, let $Z(u)$ be the plain flow just constructed and define, while $Z(u) \neq 0$,

$$c(u) = \exp\left\{-\frac{b}{a}\int_0^u \frac{dv}{\|Z(v)\|_{\mathrm{F}}}\right\}, \qquad t(u) = \frac{1}{a}\int_0^u c(v)\,dv, \qquad X(t(u)) = c(u)Z(u). \tag{17}$$

Differentiation gives $\dot{X} \in -a\partial S_k(X) - bX/\|X\|_{\mathrm{F}}$, exactly the convexified F-Fanion flow. Its extinction time is $\lim_{u \uparrow u_*^{\mathrm{P}}} t(u)$, and is at most both u_*^{P}/a and $\|X_0\|_{\mathrm{F}}/b$ when $b > 0$. The first bound uses $c \leq 1$; the second follows from $-d\|X\|_{\mathrm{F}}/dt = aS_k(X)/\|X\|_{\mathrm{F}} + b \geq b$. This statement is restricted to the isotropic quadratic and ideal continuous oracles; it is not a theorem about general Kronecker Hessians.

Flat-spectrum hybrids: exact crossover examples

If $X_0 = sQ$, with Q square orthogonal, the averaged tied maximizer $(k/r)Q$ makes the F-flow stay on the ray through Q. Hence

$$u_*^{\mathrm{P}} = \frac{rs}{k}, \qquad t_*^{\mathrm{F}} = \frac{rs}{ak + b\sqrt{r}}. \tag{18}$$

The low-rank F-family saturates at a time of order $\sqrt{r}$ instead of r/k, and changing k has little effect when $k \ll (b/a)\sqrt{r}$. This attains the nuclear-capacity lower bound for that flow.

For S-Fanions the coordinate basis matters. If Q is a normalized Hadamard matrix, so every entry is $\pm 1/\sqrt{r}$, then $\mathrm{sign}(Q) = \sqrt{r}\,Q$, and the S-flow on the same ray has

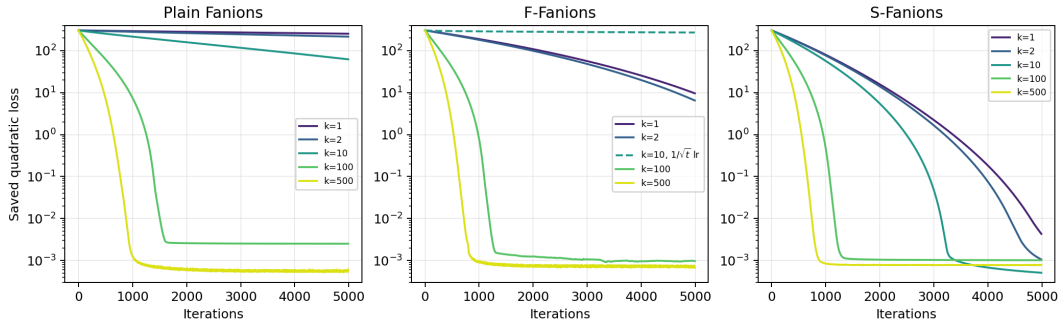
$$t_*^{\mathrm{S}} = \frac{rs}{ak + b\eta r^{3/2}}. \tag{19}$$

Such examples can be constructed at every power-of-two r using the Sylvester recursion. If instead $Q = I$, the denominator is $ak + b\eta r$. The gradient singular spectrum is identical in these two examples, but the sign term behaves differently. Entry density is therefore necessary in an explanation of the S-family. These formulas use tied averaged oracles and continuous time, as stated above.

For $r = 500$, $a = b$, and the paper's $\eta = .01$, the dense-support crossover scales are approximately 22.4 for F-Fanions and 111.8 for dense S-Fanions with maximal entry uniformity. Gaussian-like entries insert a factor $\sqrt{2/\pi}$, giving about 89.2 for the latter. These are model scales, not fitted optimum ranks or predictions of exact threshold times.

Comparison with the saved quadratic Fanions

The file `code/long_run` contains all 17 methods, 500 saved losses each, at iterations 0, 10, …, 4990. All these losses are finite. The initial loss 302.896057 is common to every method. Muon, F-Muon, and S-Muon first reach the saved 10^{-3} threshold at 1060, 910, and 890, respectively, matching the manuscript's reported values. The complete data and current-source settings are exported to `results/fanions/saved_quadratic_summary.csv`; input hashes are retained in `results/fanions/summary.json`.



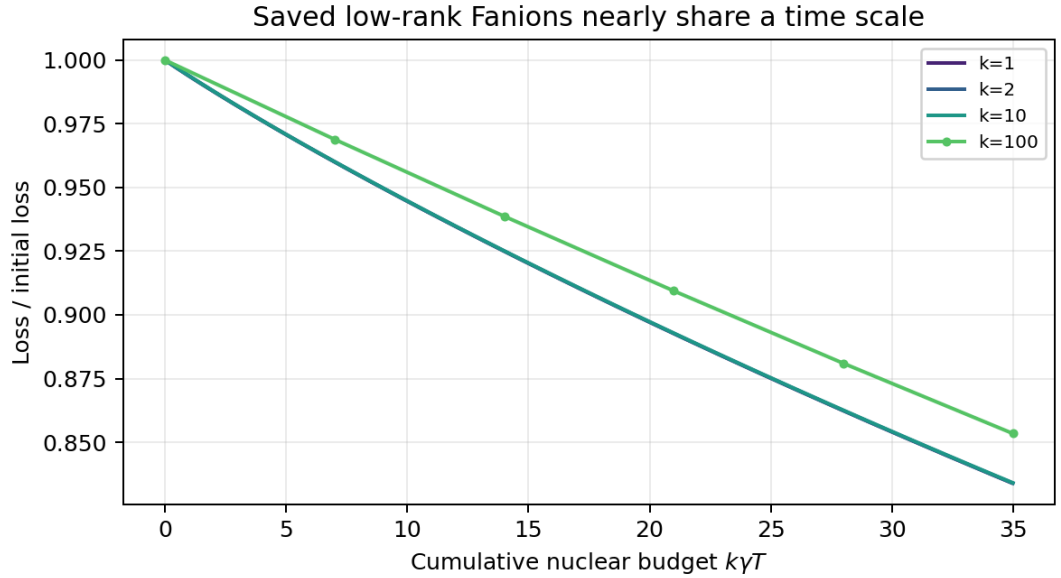
All five ranks in each saved family. The dashed F-Fanion-10 curve has a different schedule in the current source. S-Fanion-10 reaches a slightly lower saved loss floor than S-Muon; a universal ranking, at every late-time threshold, is not the experimental pattern.

The kT prediction is unusually accurate at low ranks

Compare the saved loss $f_k(t)$ with linearly interpolated Neon loss $f_1(kt)$ over their common recorded horizon, excluding $t = 0$. No horizontal scaling parameter was fitted. The maximum absolute mismatch, divided by the common initial loss, is:

Rank	Compared points	Last t	Maximum mismatch
2	249	2490	0.000252%
10	49	490	0.01884%
100	4	40	1.8583%

The rank-2 overlap is particularly striking. The rank-100 comparison has only four noninitial points and does not justify extrapolating its accuracy. These are same-problem saved traces, not comparisons to the fresh draw used for the lower-bound table. They support a low-rank budget-limited regime; they do not prove that the isotropic-flow equation governs the actual anisotropic, momentum-driven quadratic.



The plain family, plotted against $k\gamma T$ over the shared Neon horizon. Rank-100 points show the beginning of a departure.

First-order projections also distinguish the hybrids

The initial saved gradient has $\|G_0\|_{\mathrm{op}} = 2.019055$, $\|G_0\|_{\mathrm{F}} = 16.030416$, and $\|G_0\|_{*} = 250.871033$. Ignoring curvature and gradient change over the first ten updates gives $10\gamma\phi_1(G_0)$. For the sign term only, approximate $\|G_0\|_{1,\mathrm{entry}}$ by $\sqrt{2/\pi}\sqrt{d}\,\|G_0\|_{\mathrm{F}}$, the expectation for isotropic Gaussian entries at the same variance scale. This is a distributional model, not an observed entrywise norm. The comparisons are:

Method	Saved ten-step loss drop	Projection model	Relative error
Neon	0.140320	0.141334	0.723%
F-Neon	1.350525	1.353710	0.236%
S-Neon	3.606323	3.628414	0.613%

At the first step, positive scalar momentum scaling does not change any of the ideal directions. The small ten-step discrepancies are consistent with the short-interval projection model. This local agreement does not validate the Gaussian approximation or constant projected dynamics throughout training.

When should the next rank help?

At a common point with an exact gradient, let a valid directional quadratic upper model have curvature $C_k > 0$ along $D_k(G)$. Its best scalar-step decrease certificate is $S_k(G)^2/(2C_k)$. Therefore

$$\frac{S_{k+1}^2}{C_{k+1}} \geq \frac{S_k^2}{C_k} \quad \Leftrightarrow \quad \frac{C_{k+1}}{C_k} \leq \left(1 + \frac{\sigma_{k+1}}{S_k}\right)^2. \tag{20}$$

With Frobenius curvature $C_k = Lk$, put $q_k = S_k^2/(k\|G\|_{\mathrm{F}}^2)$. Then the next rank helps this certificate precisely when

$$\frac{\sigma_{k+1}}{S_k} \geq \sqrt{1 + 1/k} - 1 = \frac{1}{k(\sqrt{1 + 1/k} + 1)}. \tag{21}$$

For σ_i proportional to i^{-p} and $0 \leq p < 1$, S_k is asymptotic to a constant times k^{1-p}, so the large-k rank profile is proportional to k^{1-2p} in the range $1 \ll k \ll r$, apart from a normalizing factor independent of k. Spectra flatter than the $p = 1/2$ boundary favor more ranks in this model; faster decay eventually penalizes additional weak modes. This is a statement about the certificate's spectral term, not a universal optimizer ordering.

Nonmonotone hybrid rank curves even for a flat spectrum

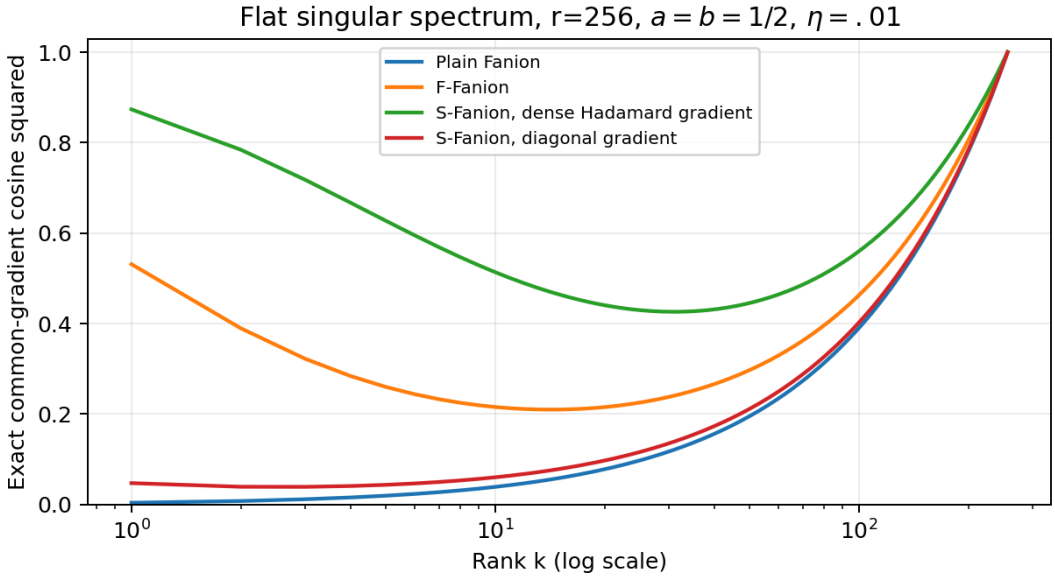
Suppose every singular value of G equals $s > 0$ and $a > 0$. For exact rank-k directions, put $t = b/a$. The F-Fanion cosine squared is

$$q_k^{\mathrm{F}} = \frac{(k/\sqrt{r} + t)^2}{k + t^2 + 2tk/\sqrt{r}}. \tag{22}$$

As a function of real $k > 0$ this decreases until $k_{\min} = tr/(\sqrt{r} + 2t)$ and increases thereafter; when $k_{\min} < 1$ the admissible ranks are on its increasing part. For $r = 64$, $a = b$, its values at $k = 1, 5, 64$ are respectively 0.5625, 0.364224, and 1. Thus a low-rank F-hybrid can beat an intermediate rank even though the full-rank hybrid has

the best score. The minimum for equal coefficients is near $\sqrt{r}$, the same scale as the dense-support crossover found earlier.

For S-Fanions at a flat Hadamard gradient, the same expression holds with $t = (b/a)\eta r$; at a flat diagonal gradient, use $t = (b/a)\eta\sqrt{r}$. These are exact common-gradient scores and do not require averaged tie choices. They illustrate how the interior rank curve changes with the hybrid and the coordinate basis. The scalar step is optimized separately in these scores; an actual fixed learning rate must instead be assessed with the next calculation.



Exact rank profiles at flat-spectrum gradients. The plain and F profiles are invariant under orthogonal changes of basis. The two S profiles have the same singular spectrum and different entry structures. All full-rank directions here are parallel to the gradient.

A fixed learning rate asks a different question.

For a quadratic with Hessian operator $\mathcal{H}$, set $P = u_{k+1}(H)v_{k+1}(H)^{\top}$. For any of the three families, let $D_{k'}$ be its direction, and let $a' = 1$ for the plain family and $a' = a$ for either hybrid. Increasing rank changes the direction by $a'P$ and its actual one-step loss reduction by

$$\gamma a'\langle G, P\rangle - \frac{\gamma^2}{2}\left(2a'\langle D_{k'}, \mathcal{H}P\rangle + (a')^2\langle P, \mathcal{H}P\rangle\right). \tag{23}$$

When $H = G$ and $\mathcal{H} = LI$, the plain criterion is simply $\sigma_{k+1} > L\gamma/2$. For the F-family it is

$$\sigma_{k+1}\left(1 - \frac{L\gamma b}{\|G\|_{\mathrm{F}}}\right) > \frac{L\gamma a}{2},$$

assuming $a > 0$ and a positive parenthesis. For the S-family the cross term also contains $\langle \mathrm{sign}(G), P\rangle$. Momentum can make $\langle G, P(H)\rangle$ differ from $\sigma_{k+1}(G)$ or even become negative. This equation identifies the exact measurements needed for every new rank.

For the manuscript's quadratic, $\mathcal{H}D = MDN$ and $C_k = \sum_{i,j\le k}(u_i^{\top}Mu_j)(v_i^{\top}Nv_j)$. The cross terms are real and need not all be positive. Assuming curvature is just k times the first-mode curvature is an additional modeling assumption.

What the norm logs can and cannot say about Fanion-5

The CIFAR logs contain only $\|G\|_{\mathrm{op}}$, $\|G\|_{\mathrm{F}}$, and $\|G\|_{*}$, rather than all singular values or the actual momentum input. They determine S_1 and S_r but not S_5. This is an identifiable mathematical limitation.

Identical recorded norms, opposite rank-5 conclusions

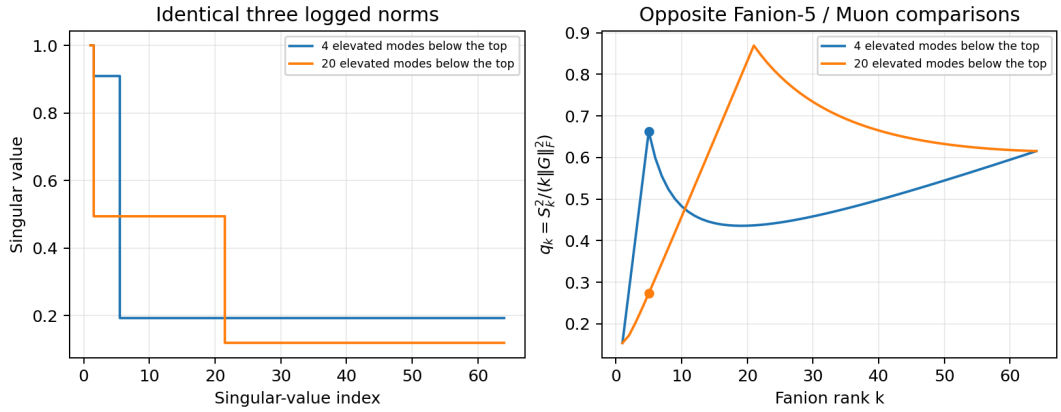
Take $r = 64$, $\sigma_1 = 1$, $\sum_i \sigma_i = 16$, and $\sum_i \sigma_i^2 = 6.5$. There are two positive spectra with these moments: one has four elevated singular values below the first, the other has twenty. They can be constructed explicitly by choosing two levels in the remaining 63 entries to match their sum and squared sum. Their scores are:

Spectrum	S_5	q_5	q_r	q_5^{F}
Four elevated modes	4.640222	0.662513	0.615385	0.824956
Twenty elevated modes	2.978569	0.272981	0.615385	0.563958

For both, $q_r^{\mathrm{F}} = 0.682593$ when $a = b = 1/2$, using

$$q_k^{\mathrm{F}} = \frac{(aS_k/\|G\|_{\mathrm{F}} + b)^2}{a^2 k + b^2 + 2abS_k/\|G\|_{\mathrm{F}}}.$$

Thus both the plain and F-family rank-5 versus full-rank comparisons reverse while all three recorded norms remain identical. These are exact Frobenius-curvature comparisons at a common gradient; any stronger inference about actual training would need more information.



The same three gradient norms admit opposite intermediate-rank comparisons. The score equals the cosine squared of the ideal update with the gradient; the best isotropic quadratic decrease is $q_k \|G\|_{\mathrm{F}}^2/2$.

Safe intervals from the existing three norms

For $1 < k < r$, write $a_0 = \|G\|_{\mathrm{op}}$, $S = \|G\|_{*}$, $Q = \|G\|_{\mathrm{F}}^2$, $n = r - 1$, $p = k - 1$, $q = r - k$, $A = S - a_0$, $V = Q - a_0^2$. Then $a_0 + L_k \le S_k \le a_0 + U_k$, where

$$h_{-} = \frac{pa_0 + A - \sqrt{(pa_0 + A)^2 - 4pV}}{2}.$$

$$L_k = \max\{pA/n,\ A - qa_0,\ 0,\ h_-\},$$
$$U_k = \min\left\{pa_0,\ A,\ \frac{pA}{n} + \sqrt{\frac{pq}{n}\left(V - \frac{A^2}{n}\right)}\right\}. \tag{24}$$

These intervals are valid but need not be the tightest possible. For their proof, put $H = \sum_{i=2}^{k} \sigma_i$. Sorting gives $H \geq pA/n$ and $H \geq A - qa_0$. Cauchy–Schwarz on the two groups gives $V \geq H^2/p + (A-H)^2/q$, yielding the upper bound. Conversely the head squares sum to at most $a_0 H$, and each tail singular value is at most H/p, so $V \leq a_0 H + H(A-H)/p$. Its smaller quadratic root gives the remaining lower bound.

For the predeclared reference file `code/cifar_norms/grad_norms_Muon_run0_20251208_162732.csv`, the six convolution layers contribute 1,500 gradient snapshots. Their ranks, inferred from the current network definition in `code/cifar_bench/norm_logger_airbench.py`, are 64 for the first two layers and 256 for the remaining four; this architecture assumption and its source hash are retained with the results. Using (24), Muon's common-point Frobenius certificate exceeds the *largest allowed* Fanion-5 certificate in 21.2% of them. The reverse is certified in 0.47%; most comparisons remain unresolved. For the F-family, F-Muon over F-Fanion-5 is certified in only 0.73%. These fractions refer to one trajectory's surrogate comparisons, not to fractions of runs where one optimizer won. The individual intervals and layerwise summaries are exported for all k in $\{1, 2, 5, 10, \min(100, r), r\}$.

Only eight of the 250 loss rows in this reference file are finite; all norm values used are finite. No loss-rate fit is made to these logs. They are also different runs from the manuscript's eight-epoch CIFAR accuracy table.

How to turn this into a stronger paper result

A defensible result already available.

Theorem 1, Proposition 3, and Theorem 4 are proved bounds for every rank and hybrid family. The exact flow and finite flat example explain when their rank dependence is attainable. The saved low-rank time comparison and the schedule audit are direct evidence specific to this repository. The short fragment `paper_insert_fanions.tex` supplies a candidate section without changing the submission.

The most useful next measurement.

At common saved model states, record the actual momentum matrix H, an independent gradient estimate G, and the actual implemented directions for each rank. Save $S_k(H)$ and $\langle G, D_k(H)\rangle$ for every desired k; three aggregate norms cannot replace them. For S-hybrids also record $\|H\|_{1,\text{entry}}$ and its sign cross terms. Estimate directional curvature and pairwise cross terms using Hessian-vector products or controlled loss probes at several step lengths. Evaluate both the frozen-state score $[\langle G, D\rangle]_+^2 / C(D)$ and the actual scheduled decrease $\gamma\langle G, D\rangle - \gamma^2 C(D)/2$.

This would test a pre-specified rank curve, including the marginal condition (23), instead of assuming that its endpoints determine the interior. Validation accuracy still needs a generalization argument; an optimization convergence bound alone does not imply an accuracy ordering.

Relation to existing literature.

The support-function/descent and best singular-approximation ingredients are standard. Crawshaw et al. develop a norm-based view that also makes layerwise aggregation and update normalization explicit. Ma, Zhang, and He analyze Schatten ball and regularization oracles with momentum, minibatching, and multiple blocks. Their reported stationarity uses the corresponding dual Schatten norm; comparing methods on a common Frobenius target requires retaining norm conversions and structural constants. Our sets C_k and their hybrid Minkowski sums make the manuscript's discrete rank parameter explicit. We make no priority claim for the general analysis or a claim that the cited work establishes a universal Fanion ordering. The first note discusses the other related Muon and structured-optimization papers.

Reproduction and numerical checks.

Run `python theory_attempts/fanion_analysis.py`, then `python theory_attempts/build_report.py --stem fanion_followup`. The analysis uses NumPy and Matplotlib on CPU. It checks 160 moment intervals, 160 comparisons to the earlier scalar barriers, 480 actual quadratic trajectory barriers, 2,482 flat finite-step identities, 160 continuous-flow support identities, 480 all-family oracle inequalities, and 400 Nesterov unrolling/tracking identities. These numerical checks support the implementation; the mathematical guarantees rest on the proofs and assumptions above. The output JSON includes source hashes and complete numbers. No new neural-network training, checkpoint recovery, or original experiment replication is claimed.

References

1. M. Crawshaw, C. Modi, M. Liu, and R. M. Gower. *An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants*. arXiv:2510.09827, 2025. https://arxiv.org/html/2510.09827.
2. B. Ma, J. Zhang, and C. He. *Schattor: Schatten-family methods for deep learning optimization*. arXiv:2606.15702, 2026. https://arxiv.org/html/2606.15702v1.